

Subject Description Form

Subject Code	DSAI4207
Subject Title	Introduction to Large Language Models
Credit Value	3
Level	4
Pre-requisite/ Co-requisite/ Exclusion	Pre-requisite: AMA1751 and COMP1011
Objectives	This course provides undergraduate students with a clear and accessible introduction to large language models (LLMs). It balances conceptual understanding with practical experience, covering how LLMs work, how they are used, and what their limitations are. Students will learn to use open-source models responsibly, explore real-world applications through hands-on exercises, and develop critical awareness of ethical and societal issues. It prepares students for advanced AI courses and real-world applications in industry or research.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a) Explain how large language models work, from basic language modeling principles to modern neural architectures. b) Use pre-trained language models responsibly to solve text-based tasks using prompting strategies and retrieval-augmented generation. c) Implement a prototype of an LLM-powered application that integrates external knowledge and evaluates output quality. d) Critically assess the limitations of large language models and discuss their ethical and societal implications.
Subject Synopsis/ Indicative Syllabus	This course introduces large language model across three key parts: Part 1: Foundations. Introduces language modeling as estimating the probability of word sequences, from n-grams to neural models. Students will learn tokenization methods, vocabulary handling, word representations, and key evaluation metrics. Part 2 Core LLMs. This part dives into how LLMs are built and trained. We'll start with the foundational Transformer architecture and compare its main variations: Encoder-only (like BERT), Decoder-only (like GPT), and Encoder-Decoder (like T5). You'll then learn practical techniques to control how the model generates text, including temperature, top-k, and top-p sampling. Finally, we'll explore the fascinating emergent abilities of LLMs, such as zero-shot reasoning, and cover the entire training lifecycle, from large-scale pretraining to crucial post-training steps like instruction tuning and alignment. Part 3: Applications. This final module turns theory into practice. Students will learn prompt engineering techniques (zero-shot, few-shot, chain-of-thought), and build a basic Retrieval-Augmented Generation (RAG) system for knowledge-grounded answers. The module also introduces adaptation methods such as prompt tuning and efficient fine-tuning, preparing students to deploy and evaluate LLM-powered systems responsibly.

Teaching/Learning Methodology	- Lectures: Conceptual explanations with visualizations, live demos, and real-world examples. Emphasis on intuition over mathematical derivation. - Tutorials / Labs: Guided Colab notebooks using Hugging Face Transformers, sentence-transformers, and FAISS. Students implement tokenization, prompting, RAG, and basic evaluation. - Course Project: Team-based (2–3 students) to design and present a simple LLM-powered application.					
Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)			
			a	b	c	d
	1. Assignments	20%	√	√	√	
	2. Course Project	20%	√	√	√	
	3. Midterm	20%	√			
	4. Final Exam	40%	√		√	√
	Total	100 %				
	Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:					
	Assignments: Tests understanding of core principles (a), practical use of models (b), and implement and evaluate (c).					
	Course Project: Assesses tool use and application development (b), model implementation and evaluation (c), and demonstration of foundational knowledge in the written report (a).					
	Midterm: Evaluates individual mastery of foundational concepts of large language models (a).					
	Final Exam: Measures comprehensive knowledge of architectures and theory (a), ability to design practical LLM-based solutions (c), and critical awareness of model limitations and societal implications (d).					
Student Study Effort Expected	Class contact:					
	▪ Lectures				26 Hrs.	
	▪ Tutorials				13 Hrs.	
	Other student study effort:					
	▪ Assignment				25 Hrs.	
	▪ Course Projects				38 Hrs.	
	▪ Exam Preparation/Self-Study/Practice				23 Hrs.	
	Total student study effort				125 Hrs.	

Reading List and References	Reference Materials: 1. Alammam, J., & Grootendorst, M. (2024). Hands-on large language models: language understanding and generation. " O'Reilly Media, Inc." 2. Tunstall, L., Von Werra, L., & Wolf, T. (2022). Natural language processing with transformers. " O'Reilly Media, Inc." 3. Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., ... & Polosukhin, I. (2017). Attention is all you need. Advances in neural information processing systems, 30. 4. Goodfellow, I., Bengio, Y., Courville, A., & Bengio, Y. (2016). Deep learning (Vol. 1, No. 2). Cambridge: MIT press. 5. Kojima, T., Gu, S. S., Reid, M., Matsuo, Y., & Iwasawa, Y. (2022). Large language models are zero-shot reasoners. Advances in neural information processing systems, 35, 22199-22213. 6. Xiao, T., & Zhu, J. (2025). Foundations of large language models. arXiv preprint arXiv:2501.09223.
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